# Effects on diabetes management of a health-care provider mediated, remote coaching system via a PDA-type glucometer and the Internet

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**Summary**

We conducted a randomized controlled trial for 12 weeks in patients with type 2 diabetes living in a rural part of Korea. The intervention group ($n = 35$) was managed by a diabetes centre which provided specialized management mediated by a primary health-care nurse who used a PDA-type blood glucometer with a bar code detector to measure the capillary glucose levels. The control group ($n = 36$) received usual care. Compared with baseline, $HbA_{1c}$ was significantly reduced at three-month follow-up in the intervention group (8.0% vs. 7.5%; $P < 0.01$), but not in the control group. Total cholesterol was significantly reduced in the intervention group (10.7 mmol/L vs. 10.4 mmol/L; $P = 0.043$). Fasting plasma glucose and triglyceride levels were lower at follow-up in both groups, but the difference was not significant. The new system could be implemented widely and would contribute to improving the quality of diabetes care, even in isolated rural areas.

## Introduction

It is well known that appropriate glucose control can reduce the risk of developing diabetes complications in people with type 1 and type 2 diabetes.[1–3] Intensive monitoring, continuing patient motivation and education by an integrated diabetes care team are the essential components for achieving treatment goals in patients with diabetes.[4–6] However, medical resources including specialized physicians, diabetes educators and education programmes are rarely available in a primary health-care setting in rural areas. Thus, there is a need to improve support for patients with diabetes in these areas.

Korea has a well-organized public health-care network that covers the whole nation. There are 251 public health-care centres, one in each district of the cities and one in each county of the rural areas. Public health-care centres located in rural areas are linked with 10–15 branches, so that there are about 1300 health-care branch offices.[7] These are operated by one general physician and several nurses and provide basic health care to the inhabitants. In addition, many of these branches link with one to three health-care posts located in isolated rural areas. There are 1911 of these health-care posts staffed by a single nurse caregiver. At each health-care post the nurse caregiver monitors and treats 20–50 diabetes patients. Lack of special education about diabetes management for these nurse caregivers and limited care resources are the main barriers to achieving the desired treatment goals in these patients.

In a previous report, we described a bidirectional communication system between patients and doctors using the Internet and demonstrated its short- and long-term effects on glucose control.[8–9] Other diabetes management systems using various information technologies have also shown beneficial effects in the management of diabetes.[10–18] Many systems use direct communication between doctors and patients, but ours focused on nurse caregivers in rural area as the mediator between doctors and patients. Recently, we have developed a new health-care support system based on Internet communication. This allows diabetes care resources, including education programmes about diet and exercise for diabetic patients, to be delivered to nurse caregivers at health-care posts. The nurse caregiver can also communicate through the Internet with the diabetes care team in our institution to solve individual patient problems. In addition, the nurse caregiver can use PDA-type blood glucometers with a bar code detector to measure the capillary glucose levels of patients and also discuss with each patient the data previously stored in the PDA. All the data recorded in the PDA are automatically transmitted to the diabetes centre when the PDA is connected to a computer.

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The primary objective was to test whether $HbA_{1c}$ improved after three months in a controlled trial.

## Methods

A randomized controlled design was used. We selected six socioeconomically and demographically similar public health-care posts, associated with Chung-ju city (about 150 km from the main city Seoul), by simple random selection. We invited patients who were registered with the primary health-care posts to participate. The inclusion criteria were: a diagnosis of type 2 diabetes; patients aged $\geq$40 years and with an $HbA_{1c}$ of 7.0–11.0% who had been followed for more than six months by the health-care post. Exclusion criteria were: (1) diagnosed or suspected disease of the liver, pancreas, endocrine organs or kidneys; (2) ischaemic heart disease or cerebrovascular disease or a history of such disease; (3) a creatinine higher than 0.133 mmol/l; (4) treatment with intensive insulin regimens or (5) those unable to attend regularly. All patients provided written informed consent and the study was approved by the appropriate ethics committee.

### Intervention

The nurses in the primary health-care posts received training at the Chung-ju Health centre in delivery of the intervention. For the intervention group, the nurses at each primary health-care post measured patients' glucose levels with a PDA-type blood glucometer (Accu-Check Inform System, Roche) after identifying the subject with the PDA barcode detector.

After measuring glucose levels, the nurse sent the measured glucose data to the diabetes centre at the St Mary's Hospital in Seoul. The nurse also uploaded patient information such as diet, exercise and medications. A physician at the diabetes centre analysed the information and then sent the recommendations for the patients once a week via the high-speed Internet line (intensive feedback) and replied to their questions within 24 hours. The nurse contacted the patients and answered any questions. If nurses in the primary health-care posts needed information about individualized diet and exercise prescriptions for the patients, the diabetes educators in the centre sent the appropriate prescriptions and education materials. The nurse at the health office could then educate the patient face-to-face according to the physician's recommendations (Figure 1).

### Control group

The nurses at each primary health-care post also measured the control group patients' glucose levels with the PDA-type glucometer. In contrast to the intervention group, they provided general diabetes education for the subjects of the control group but the measured glucose data were not sent to the diabetes centre, so physician recommendations were not provided (Figure 1).

### Measures

At the initial interview, bodyweight, height, waist circumference and blood pressure were checked. After an

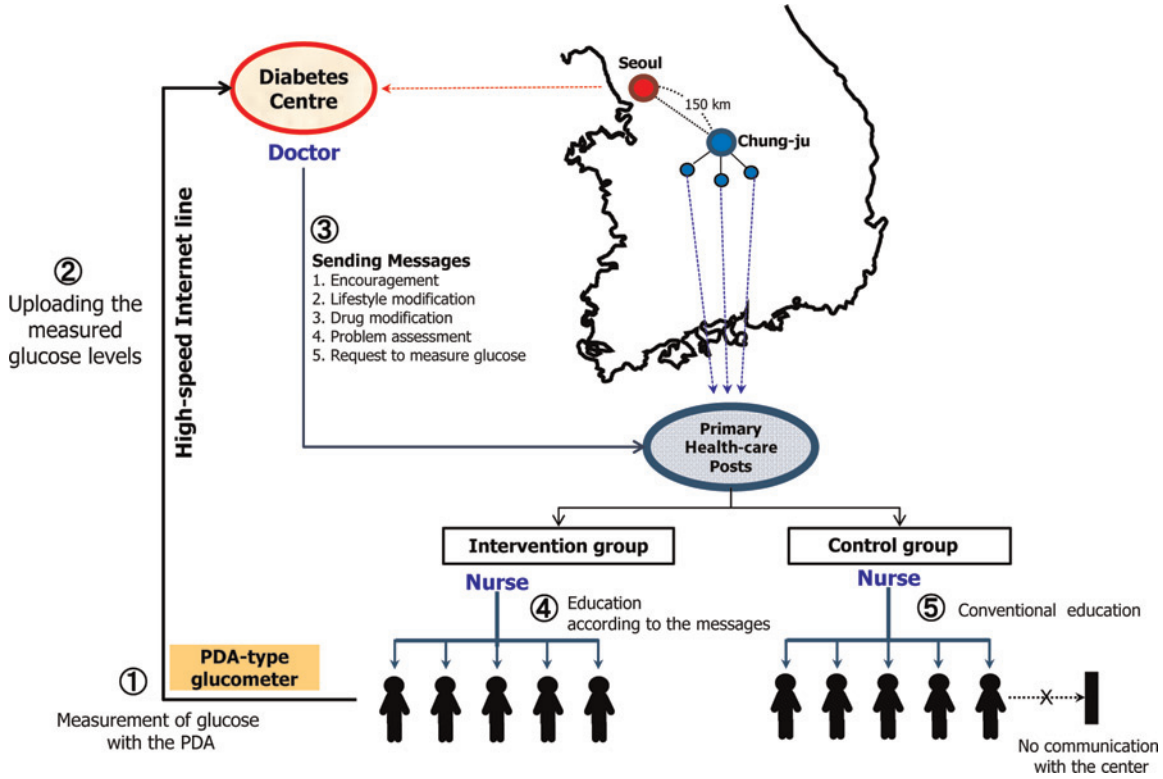


**Figure 1** Study design, using the Internet-based diabetes management system

overnight fast, blood was collected at baseline and at 12-weeks follow-up for the assessment of $HbA_{1c}$, fasting plasma glucose, lipids (total cholesterol, triglyceride, low density lipoprotein cholesterol and high density lipoprotein cholesterol), liver enzymes and creatinine.

### Statistical analysis

By assuming a SD of 1% [6], with a two-sided significance level of 0.05, there was a power of 0.80 for detecting a mean change of 0.75% in $HbA_{1c}$ from baseline to post-intervention and a power of 0.80 for detecting a difference of 0.7% in the mean change in $HbA_{1c}$ between the two study groups (each $n = 32$). We assumed a dropout rate of 10% from the study.

Baseline characteristics were compared between groups using $c^2$ tests of independence and *t*-tests for continuous variables, which were first assessed for normality. We examined the difference between groups in the mean individual changes in $HbA_{1c}$ from randomization to the end of the trial using an unpaired *t*-test. We also conducted an analysis of the within-group mean change from baseline to the end of the trial using paired *t*-tests. Unless otherwise noted, all analyses were conducted by applying intention-to-treat methods with baseline values carried forward for missing values. Data were analysed using a standard package (SAS version 8.2).

## Results

Initially, 115 individuals with type 2 diabetes were screened by a review of their medical records at the health-care posts. Thirteen subjects were not eligible because of their age, follow-up duration or $HbA_{1c}$ results. A further 31 were excluded because 18 did not attend, 10 had exclusion diseases, one was treated with insulin and two received education for diabetes mellitus from other web sites (Figure 2). A total of 71 subjects were included between July and August 2005. Thirty-six were assigned to the intervention group and 35 to the control group using a table of random numbers. At three months, the overall dropout rates were similar in both groups: three and four subjects in the control and intervention groups, respectively, missed the follow-up test.

The baseline characteristics of the two groups were similar except for blood pressure (Table 1). Eight of 35 subjects (23%) in the control group and seven of 36 subjects (19%) in the intervention group managed their diabetes with lifestyle modification, while 27 (77%) and 29 (81%) patients in the control and intervention groups, respectively, used oral antidiabetic medication. There was no use of insulin. Hypertension was present in 34% (12/35) of the control group and 47% (17/36) of the intervention group.

### Glucose control

The comparison of $HbA_{1c}$ between baseline and follow-up showed that $HbA_{1c}$ was significantly reduced from 8.0% (SD 0.8) to 7.5% (SD 0.9) in the intervention group ($P < 0.01$) and from 8.0% (SD 1.0) to 7.8% (SD 1.1) in the control group ($P = 0.11$) (Figure 3a). The mean change in $HbA_{1c}$ from baseline was 0.2% in the control group and 0.5% in the intervention group. Fasting plasma glucose level showed a tendency to decrease between baseline and follow-up in both groups, but there was no significant difference (from baseline 8.9 mmol/L (SD 1.9) to 8.4 mmol/L (SD 2.7) in the control group, $P = 0.34$; from baseline 7.9 mmol/L (SD 2.2)

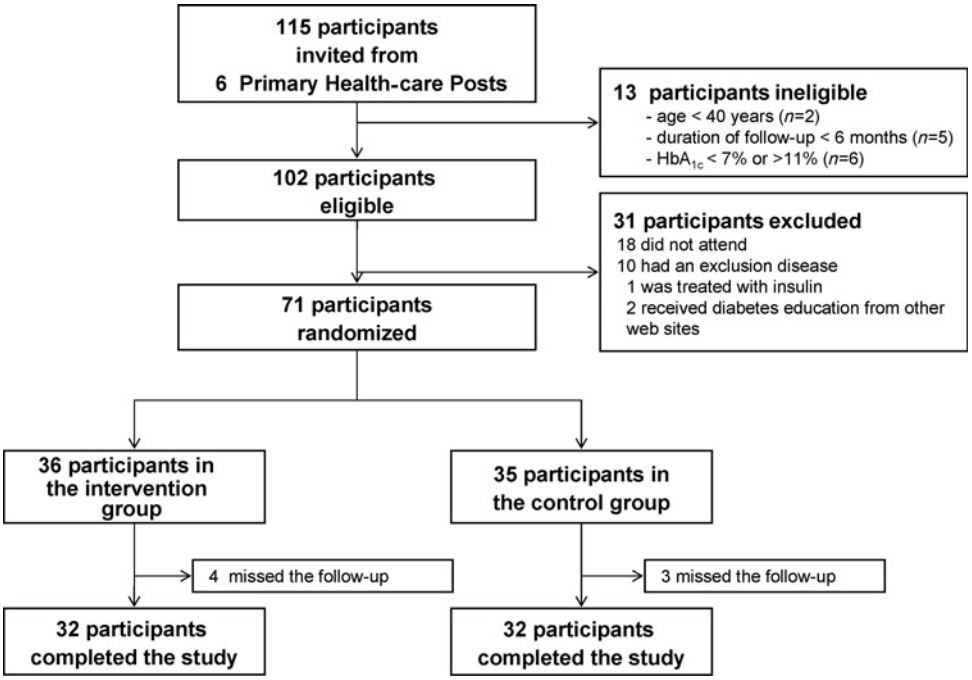


**Figure 2** Flow diagram of the study

**Table 1** Clinical characteristics of the subjects. Values are means (SD)

| | Control group | Intervention group | *P* value |
|---|---|---|---|
| No of subjects | 35 | 36 | |
| Proportion of men, % | 34 | 44 | |
| Age, years | 63.1 (10.3) | 65.3 (9.3) | 0.34 |
| BMI, kg/m$^2$ | 24.7 (3.1) | 25.2 (3.4) | 0.50 |
| Waist circumference, cm | 87.2 (7.9) | 84.3 (12.2) | 0.26 |
| Duration of diabetes, years | 9.9 (9.6) | 7.9 (6.8) | 0.31 |
| Blood pressure, mm Hg | | | |
| systolic | 120.2 (14.7) | 134.8 (16.5) | <0.001 |
| diastolic | 76.5 (7.6) | 82.9 (9.1) | 0.002 |
| Fasting plasma glucose, mmol/L | 8.9 (1.9) | 7.9 (2.2) | 0.062 |
| $HbA_{1c}$, % | 8.0 (1.0) | 8.0 (0.8) | 0.78 |
| Lipids, mmol/L | | | |
| total cholesterol | 4.8 (0.8) | 5.0 (0.9) | 0.54 |
| triglycerides | 2.7 (2.4) | 2.7 (2.0) | 0.99 |
| HDL-cholesterol | 1.2 (0.3) | 1.2 (0.4) | 0.99 |
| Glucose control methods | | | 0.73 |
| Proportion using lifestyle modification, % | 23 | 19 | |
| Proportion using oral medication, % | 77 | 81 | |

to 7.6 mmol/L (SD 3.0) in the intervention group, $P = 0.62$) (Figure 3b). Total cholesterol was significantly decreased in the intervention group (from 5.0mmol/L (SD 0.9) to 4.8 mmol/L (SD 0.7), $P = 0.043$) whereas the control group showed a tendency to increased total cholesterol (from 4.9 mmol/L (SD 0.8) to 5.0 mmol/L (SD 0.8), $P = 0.22$) (Figure 3c). Triglyceride tended to decrease in both groups (2.7 mmol/L (SD 2.4) vs. 2.4 mmol/L (SD 1.2) in the control group, $P = 0.35$; 2.7 mmol/L (SD 2.0) vs. 2.2 mmol/L (SD 1.4) in the intervention group, $P = 0.19$) (Figure 3d). There was no difference between baseline and follow-up in HDL-cholesterol, or liver enzymes in either group.

## Self-monitoring blood glucose and recommendations

The mean number of times that self-monitoring blood glucose (SMBG) was performed was 10.8 /person (SD 5.4) in the control group and 17.0 /person (SD 5.7) in the intervention group ($P < 0.01$).

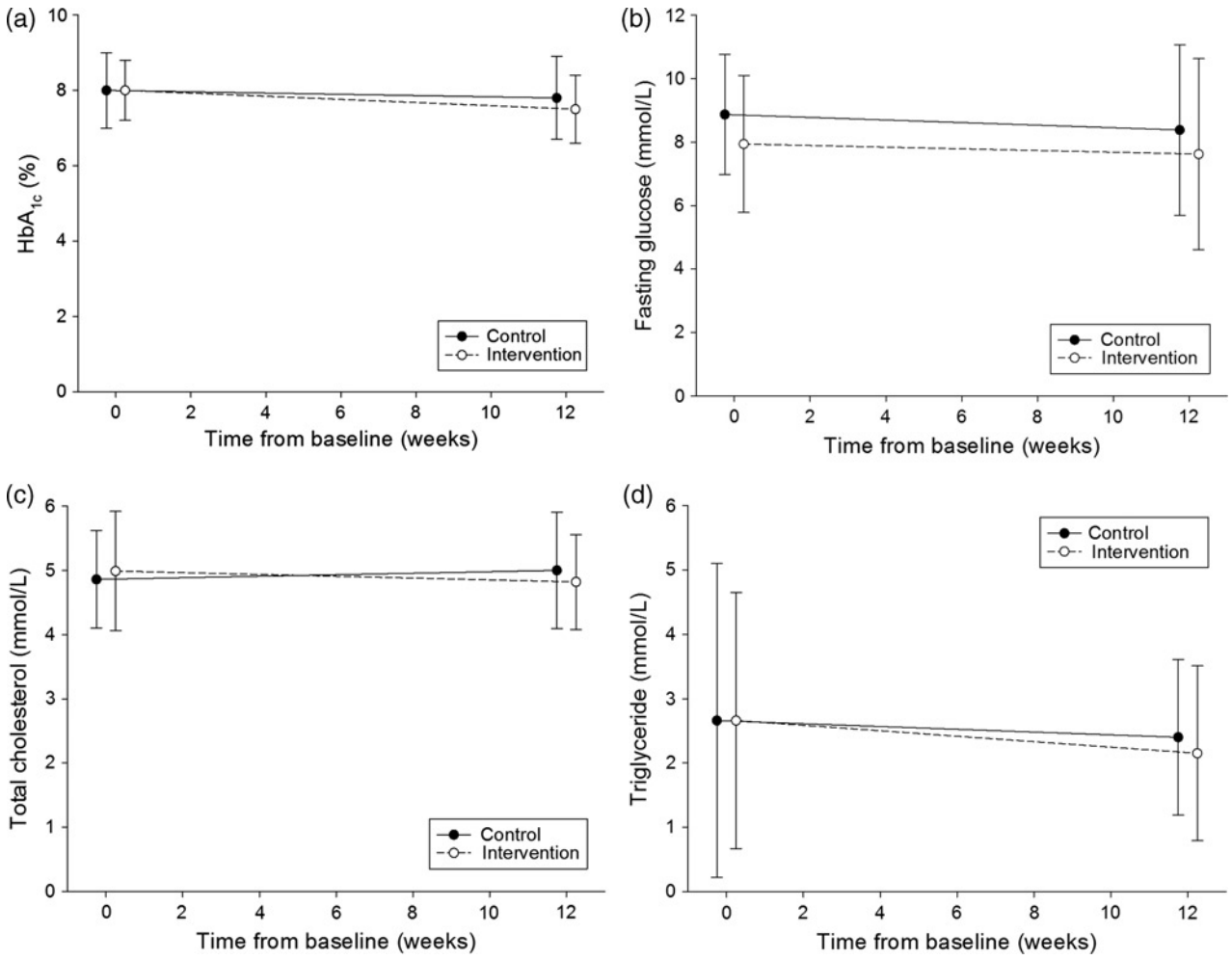


**Figure 3** Difference in outcomes in the two groups. a, $HbA_{1c}$ – this was significantly reduced in the intervention group ($P < 0.01$); b, fasting glucose; c, total cholesterol – this was significantly reduced in the intervention group ($P < 0.05$); d, triglyceride

**Table 2** Physicians' recommendations made via the online system

| | Number of recommendations (%) |
|---|---|
| Encouragement | 131 (49) |
| Lifestyle modification | 30 (11) |
| Drug modification | 31 (12) |
| Problem assessment and counselling | 37 (14) |
| Request to measure blood glucose | 40 (15) |
| *Total* | *269* |

For the intervention group, the number of recommendations sent from the diabetes centre to the primary health-care post in the rural area was 269 (mean 8.4/person) during the study period of 12 weeks. The contents of recommendations were categorized as 'encouragement', 'lifestyle modification', 'drug modification', 'problem assessment and counselling' and 'request to measure blood glucose'.

Of the recommendations sent by a physician, most (131 of 269 [49%]) were in the form of 'encouragement' (e.g.'your glucose control is good'). The frequency of 'drug modification' (e.g. 'your drug modification is needed') was 31 of 269 (12%). Most drug modifications were sent in the first four-week period (Table 2).

## Discussions

In a previous study, we described a diabetes management system using the Internet and demonstrated its short- and long-term effect on glucose control in type 2 diabetes.[8,9] However, to use the system, patients had to visit the diabetes centre periodically and also record their glucose data themselves at home on a web chart accessed through the Internet. This meant that the system was most useful in urban areas with good Internet access and where a specialized diabetes centre was close by. It was not easily applicable to rural areas because of the lack of Internet service in many houses and the number of elderly patients who were not familiar with the Internet.

The new diabetes care delivery system is used in a four-step process. The first step is measuring patients' blood glucose with the PDA-type blood glucometer. The second step is to send the glucose data directly to the diabetes centre by connecting the PDA to a PC. In the third step, physicians working in the diabetes centre review the recorded glucose data and patient information and send their opinion to the nurse at the public health post. Last, the nurse educates the patients according to the physician's recommendations (Figure 1). With this system, we can deliver specialized diabetes management to people in an isolated rural area without a face-to-face interview with a physician.

The present study showed that the mean $HbA_{1c}$ in the intervention group decreased significantly. Fasting plasma glucose showed a tendency to decrease in both intervention and control groups, although this was not significant. Using the system, we provided 269 recommendations to the intervention group, of which 31 (12%) were categorized as 'drug modification'. Such action is only possible in the new system. In addition, the frequency of blood glucose measurement was significantly higher in the intervention group than in the control group, which suggests that the system may motivate the nurses as well as the patients. This increased frequency of blood glucose measurement might also contribute to the observed improvement in $HbA_{1c}$.

In the study, we tried to develop a more practical health-care delivery system that could be applied to the general population. We expect that this management system could be used in countries such as the UK, where public health-care networks are well developed, and would also be very effective for countries such as the US and China that have a very large land area. It could also be effective for people who live on islands. The system is likely to work better for patients with type 2 diabetes than those who require intensive insulin treatment. However, there are still some problems to resolve before implementing the system for the general population, notably the establishment of IT infrastructure and public health-care resources.

In conclusion, we have developed a new diabetes care delivery system that was used with the pre-existing public health-care network and have confirmed its efficacy in treating type 2 diabetes. The new system could be implemented widely and would contribute to reducing diabetic complications and improving the quality of life for patients with diabetes, even in isolated rural areas.

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